GEV Hazard Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — GEV Hazard Model (MKM-GH-001)

Tests run on **29 March 2026 at 10:29:14**.

Table 1: Test Results Summary — GEV Hazard Model (MKM-GH-001)

Total Tests	16
Passed	16
Failed	0
Skipped	0
Duration	0.27s

Table 2: Detailed Test Results — GEV Hazard Model

Test Description	Acceptance Criteria	Result	Time
Exceedance probability in range	`0.0 <= prob <= 1.0`	Pass	0.001s
Fit returns three floats	`isinstance(shape, float)`; `isinstance(loc, float)` (+1 more)	Pass	0.013s
Force gumbel sets shape zero	`shape == 0.0`	Pass	0.000s
Higher threshold lower probability	`p_high < p_low`	Pass	0.000s
Return level increases with period	`rl_100 > rl_10`	Pass	0.000s
Compute all responses returns dict	`isinstance(responses, dict)`; `len(responses) == len(synthetic_flat_gauges)` (+1 more)	Pass	0.000s
Compute response returns gauge response	`isinstance(response, GaugeResponse)`; `response.gauge_id == gauge_id` (+1 more)	Pass	0.000s
Response threshold flags	`response.exceeded_alert`; `response.exceeded_warning`	Pass	0.000s
Annual flood probabilities in range	`0.0 <= curve.annual_flood_prob_alert <= 1.0`; `0.0 <= curve.annual_flood_prob_warning <= 1.0` (+1 more)	Pass	0.083s
Build returns dict of curves	`isinstance(curves, dict)`; `len(curves) == len(synthetic_flat_gauges)` (+2 more)	Pass	0.080s
Each curve is gauge hazard curve	`isinstance(curve, GaugeHazardCurve)`; `curve.gauge_id == gauge_id`	Pass	0.088s
Cumulative prob increasing	`probs[i] >= probs[i - 1]`	Pass	0.000s
Probabilities sum to one	`abs(point.survival_prob + point.prob_at_least_one - 1.0) ...`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Returns correct length	`len(ts) == 5`	Pass	0.000s
Returns term structure points	`isinstance(point, TermStructurePoint)`	Pass	0.000s
Survival prob decreasing	`survivals[i] <= survivals[i - 1]`	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (16 tests)	D. Kelly
29-Mar-2026	1.1	16 test(s) added; 16 test(s) removed (total: 32)	D. Kelly